ISWE406P

Agile Development Process and DevOps Lab

Assessment-1

Faculty: Dr. Sweta Bhattacharya
Slot: L3+L4

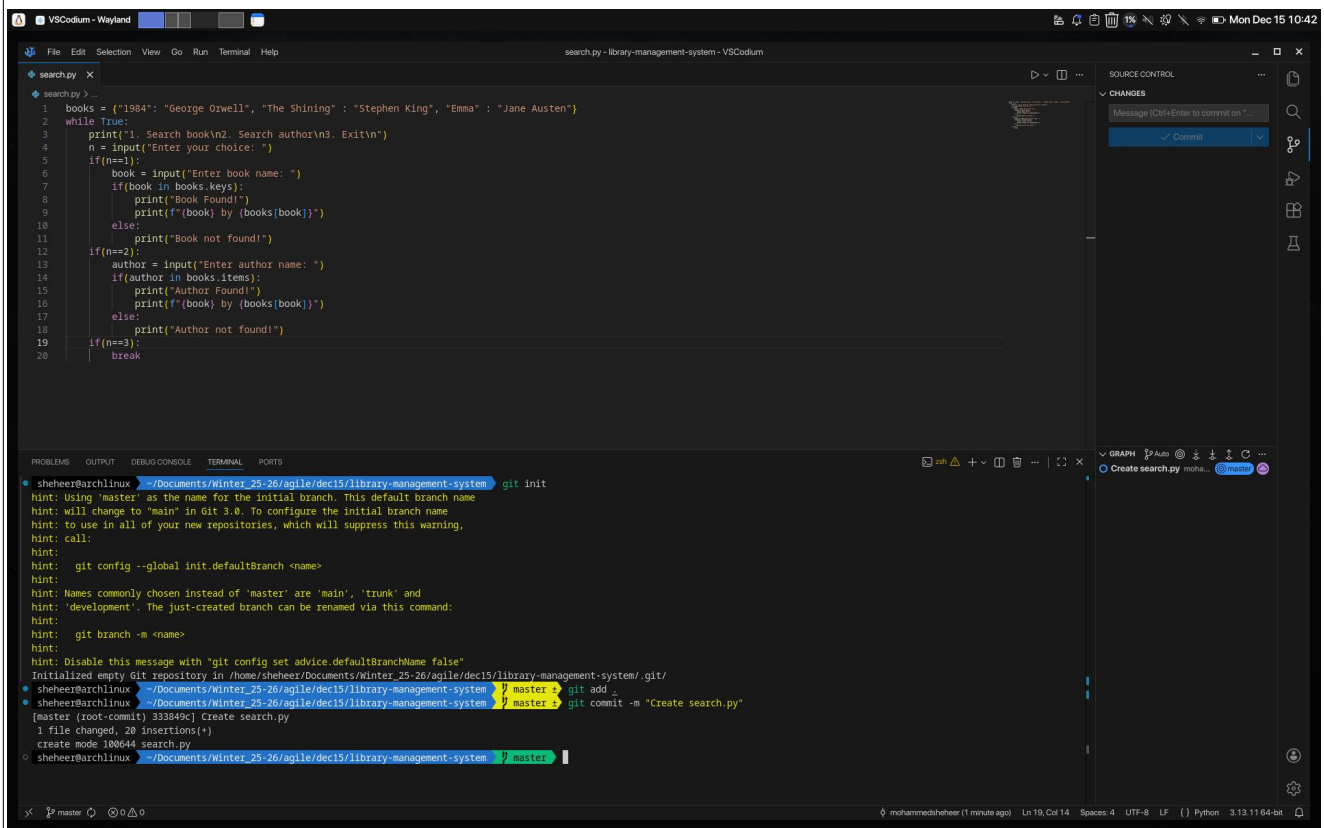
Name: Mohammed Sheheer K U
Reg No: 23MIS0170

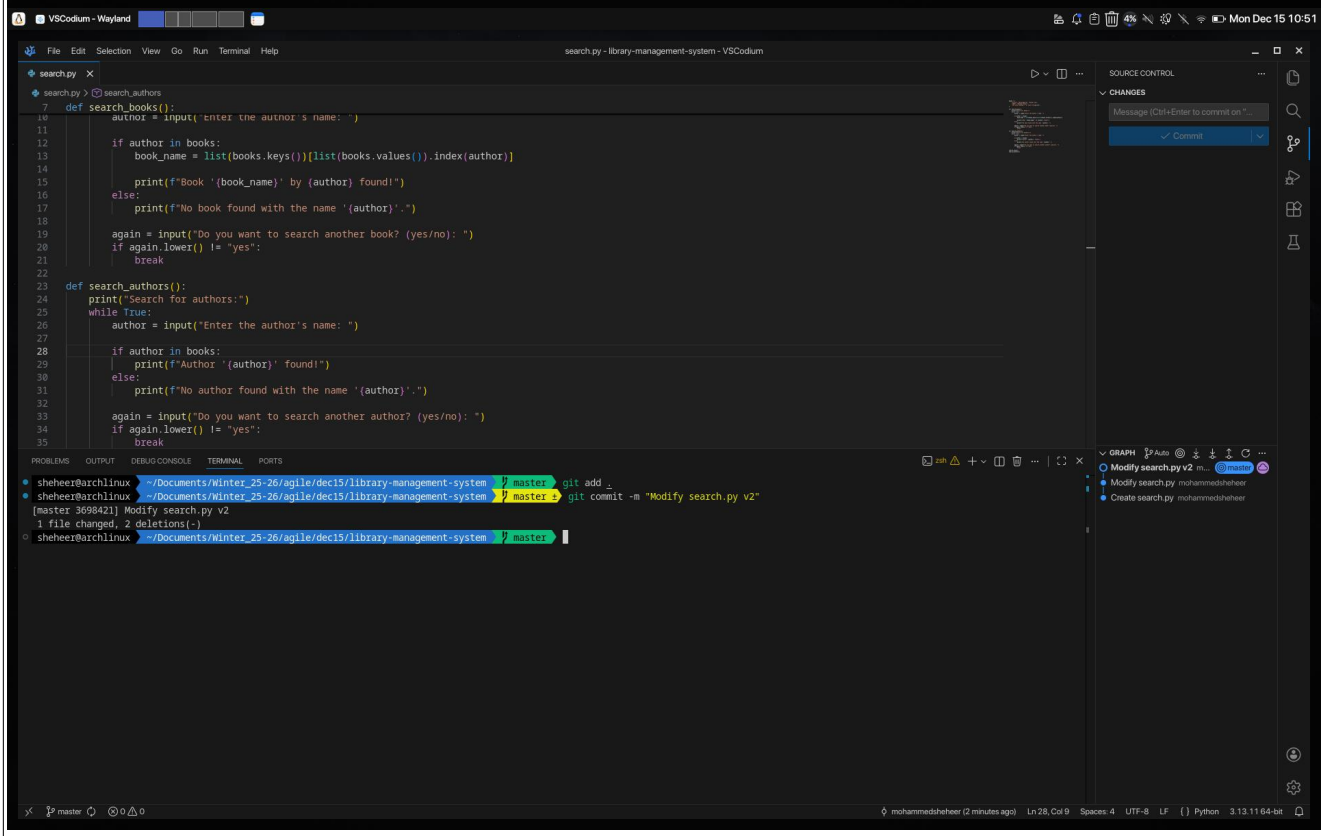
Home Work

Case Studies

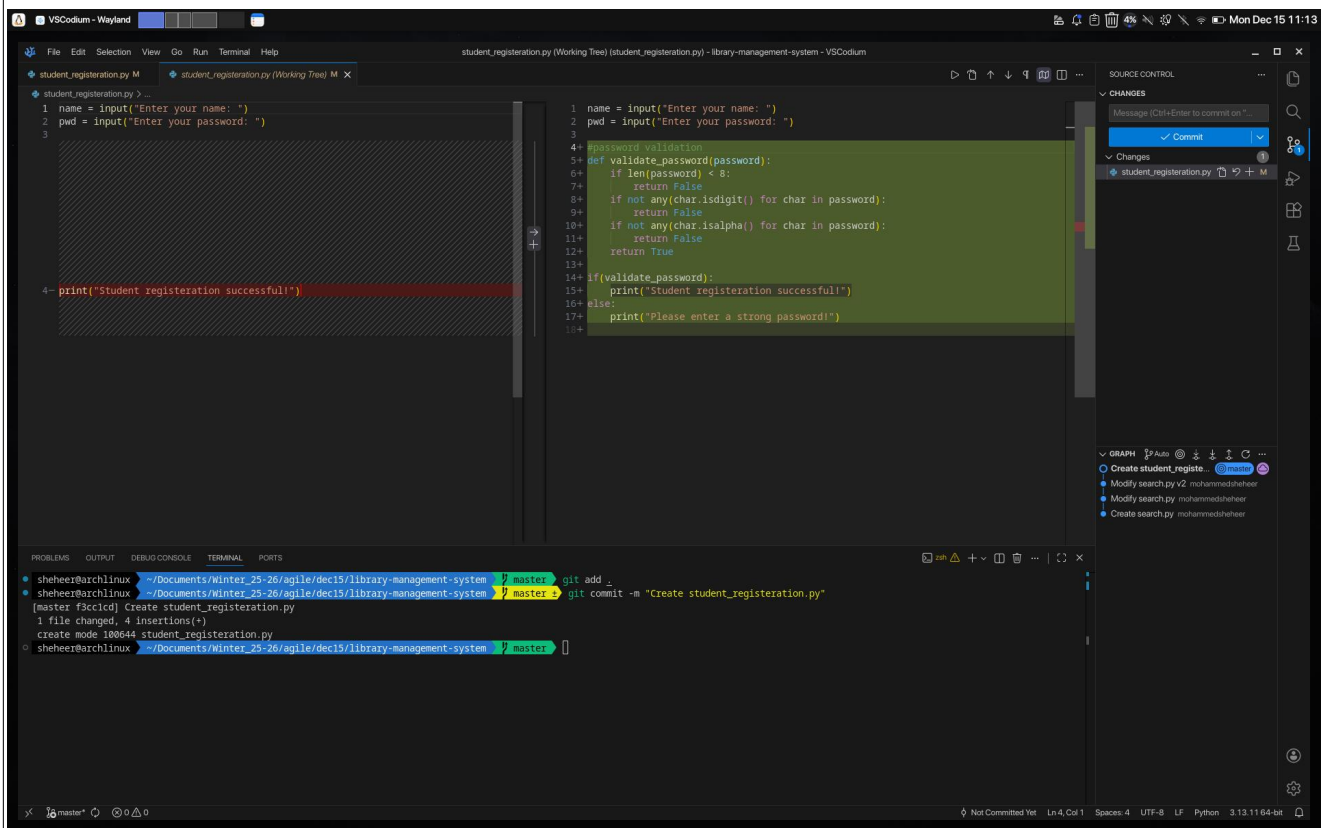
*Use appropriate **Git** Commands to Execute the following case studies in **VS Codium***

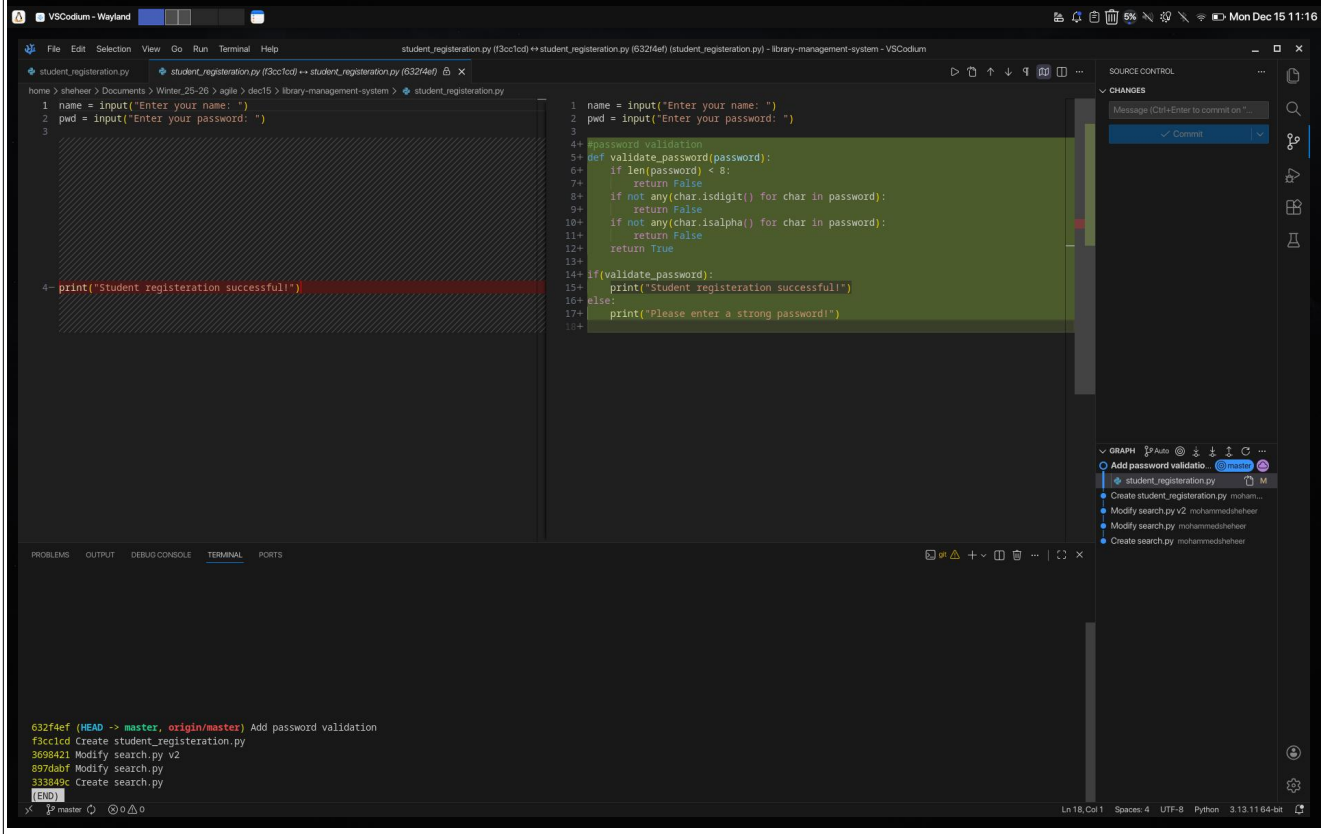
Case Study 1: New Project Version Control Setup





Case Study 2: Tracking Daily Code Changes





VSCodium - Wayland Mon Dec 15 11:16

File Edit Selection View Go Run Terminal Help student_registeration.py (f3cc1cd) ↔ student_registeration.py (632f4ef) (student_registeration.py) - library-management-system - VSCodium

student_registeration.py | student_registeration.py (f3cc1cd) ↔ student_registeration.py (632f4ef)

home > sheheer > Documents > Winter_25-26 > agile > dec15 > library-management-system > student_registeration.py

```
1  name = input("Enter your name: ")
2  pwd = input("Enter your password: ")
3
4- print("Student registeration successful!")
```

```
1   name = input("Enter your name: ")
2   pwd = input("Enter your password: ")
3
4+  #password validation
5+  def validate_password(password):
6+      if len(password) < 8:
7+          return False
8+      if not any(char.isdigit() for char in password):
9+          return False
10+     if not any(char.isalpha() for char in password):
11+         return False
12+     return True
13+
14+ if(validate_password):
15+     print("Student registeration successful!")
16+ else:
17+     print("Please enter a strong password!")
18+
```

SOURCE CONTROL
CHANGES
Message (Ctrl+Enter to commit on "...
Commit

GRAPH Auto
Add password validatio... master
student_registeration.py M
Create student_registeration.py moham...
Modify search.py v2 mohammedsheheer
Modify search.py mohammedsheheer
Create search.py mohammedsheheer

PROBLEMS OUTPUT DEBUG CONSOLE TERMINAL PORTS

```
sheheer@archlinux ~/Documents/Winter_25-26/agile/dec15/library-management-system  master  git add .
sheheer@archlinux ~/Documents/Winter_25-26/agile/dec15/library-management-system  master ±  git commit -m "Create student_registeration.py"
[master f3cc1cd] Create student_registeration.py
 1 file changed, 4 insertions(+)
 create mode 100644 student_registeration.py
sheheer@archlinux ~/Documents/Winter_25-26/agile/dec15/library-management-system  master  git add student_registeration.py
sheheer@archlinux ~/Documents/Winter_25-26/agile/dec15/library-management-system  master ±  git commit -m "Add password validation"
[master 632f4ef] Add password validation
 1 file changed, 14 insertions(+), 1 deletion(-)
sheheer@archlinux ~/Documents/Winter_25-26/agile/dec15/library-management-system  master  git log --oneline
sheheer@archlinux ~/Documents/Winter_25-26/agile/dec15/library-management-system  master
```

master 0 0 Ln 18, Col 1 Spaces: 4 UTF-8 Python 3.13.11 64-bit

VSCodium - Wayland Mon Dec 15 11:17

File Edit Selection View Go Run Terminal Help student_registeration.py (f3cc1cd) ↔ student_registeration.py (632f4ef) (student_registeration.py) - library-management-system - VSCodium

student_registeration.py | student_registeration.py (f3cc1cd) ↔ student_registeration.py (632f4ef)

home > sheheer > Documents > Winter_25-26 > agile > dec15 > library-management-system > student_registeration.py

```
1  name = input("Enter your name: ")
2  pwd = input("Enter your password: ")
3
```

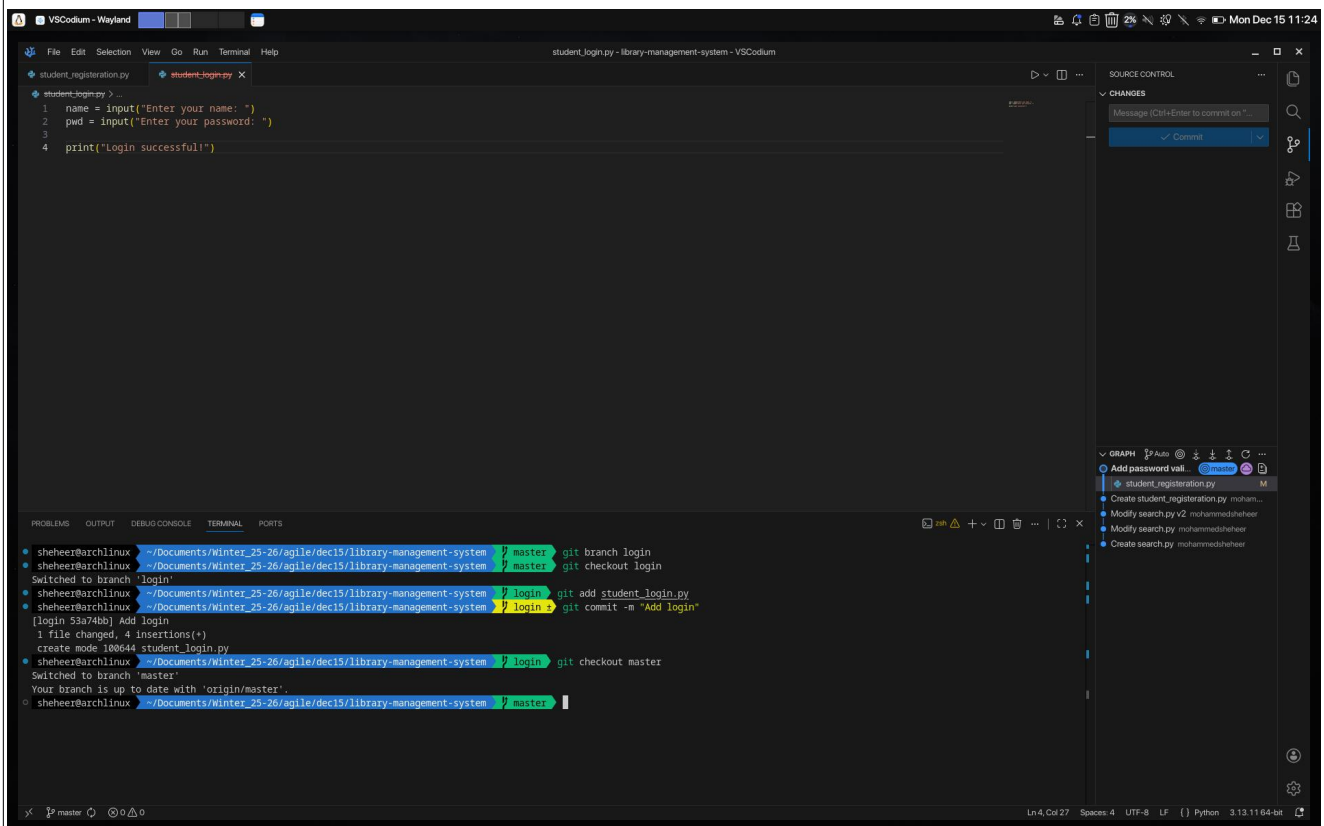
```
1   name = input("Enter your name: ")
2   pwd = input("Enter your password: ")
3
4+  #password validation
5+  def validate_password(password):
6+      if len(password) < 8:
7+          return False
8+      if not any(char.isdigit() for char in password):
9+          return False
10+     if not any(char.isalpha() for char in password):
```

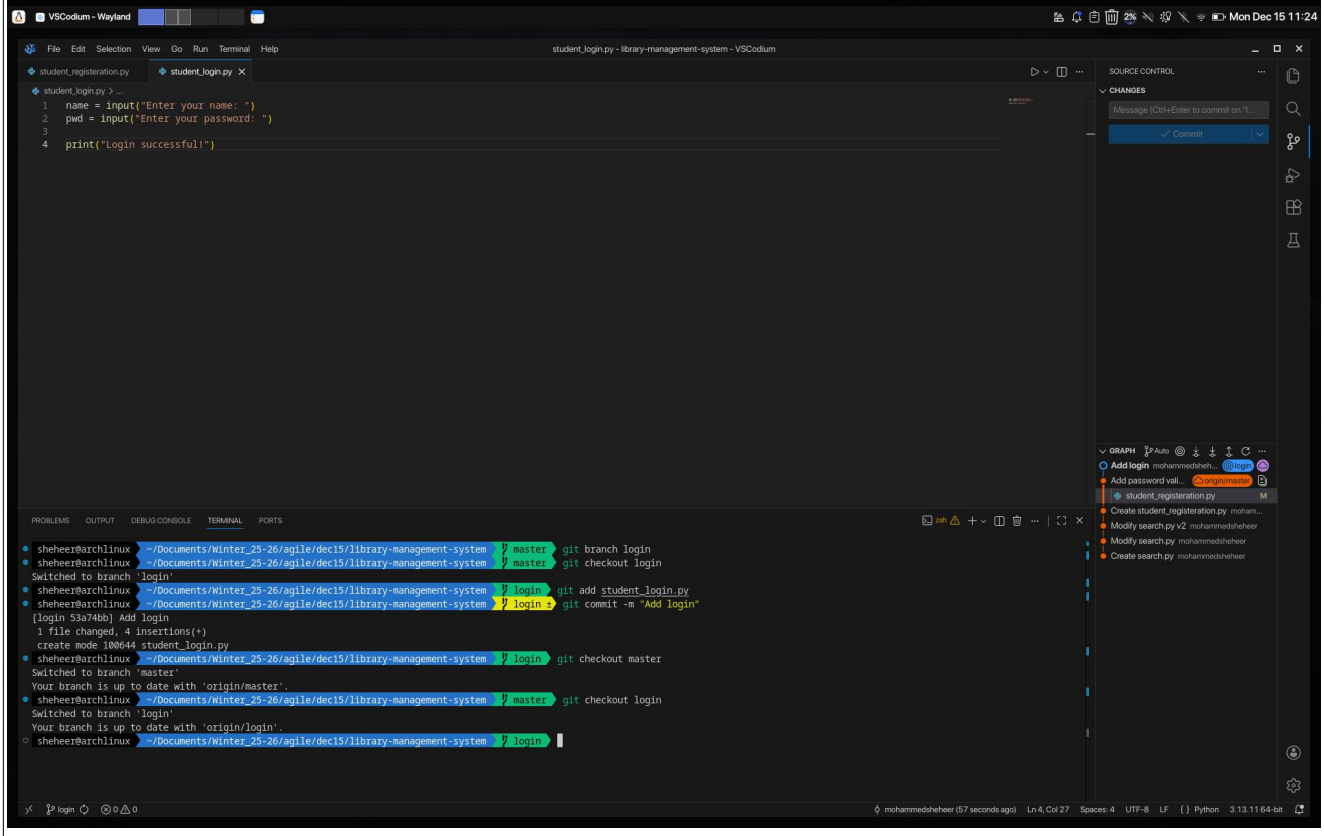
PROBLEMS OUTPUT DEBUG CONSOLE TERMINAL PORTS

```
commit 632f4efffe982b8fabee73156f48542f55d9f575 (HEAD -> master, origin/master)
Author: mohammedsheheer <mohammed.sheheer2@gmail.com>
Date:   Mon Dec 15 11:15:51 2025 +0530

    Add password validation

commit f3cc1cd71c1fd0feb87fb4c2c4fe84eec7e5cc4a
Author: mohammedsheheer <mohammed.sheheer2@gmail.com>
Date:   Mon Dec 15 11:04:41 2025 +0530

    Create student_registeration.py

commit 36984215021bffe23f44c3e351c1de7ca62f5ba6
Author: mohammedsheheer <mohammed.sheheer2@gmail.com>
Date:   Mon Dec 15 10:50:51 2025 +0530

    Modify search.py v2

commit 897dabf744822b7dcc4bf55313d0a9ef377bd1a7
Author: mohammedsheheer <mohammed.sheheer2@gmail.com>
Date:   Mon Dec 15 10:48:42 2025 +0530

    Modify search.py

commit 333849c6a42075713a0f4014807a1051b761f265
Author: mohammedsheheer <mohammed.sheheer2@gmail.com>
Date:   Mon Dec 15 10:39:58 2025 +0530

    Create search.py
~
~
~
~
~
~
~
~
~
(END)
```

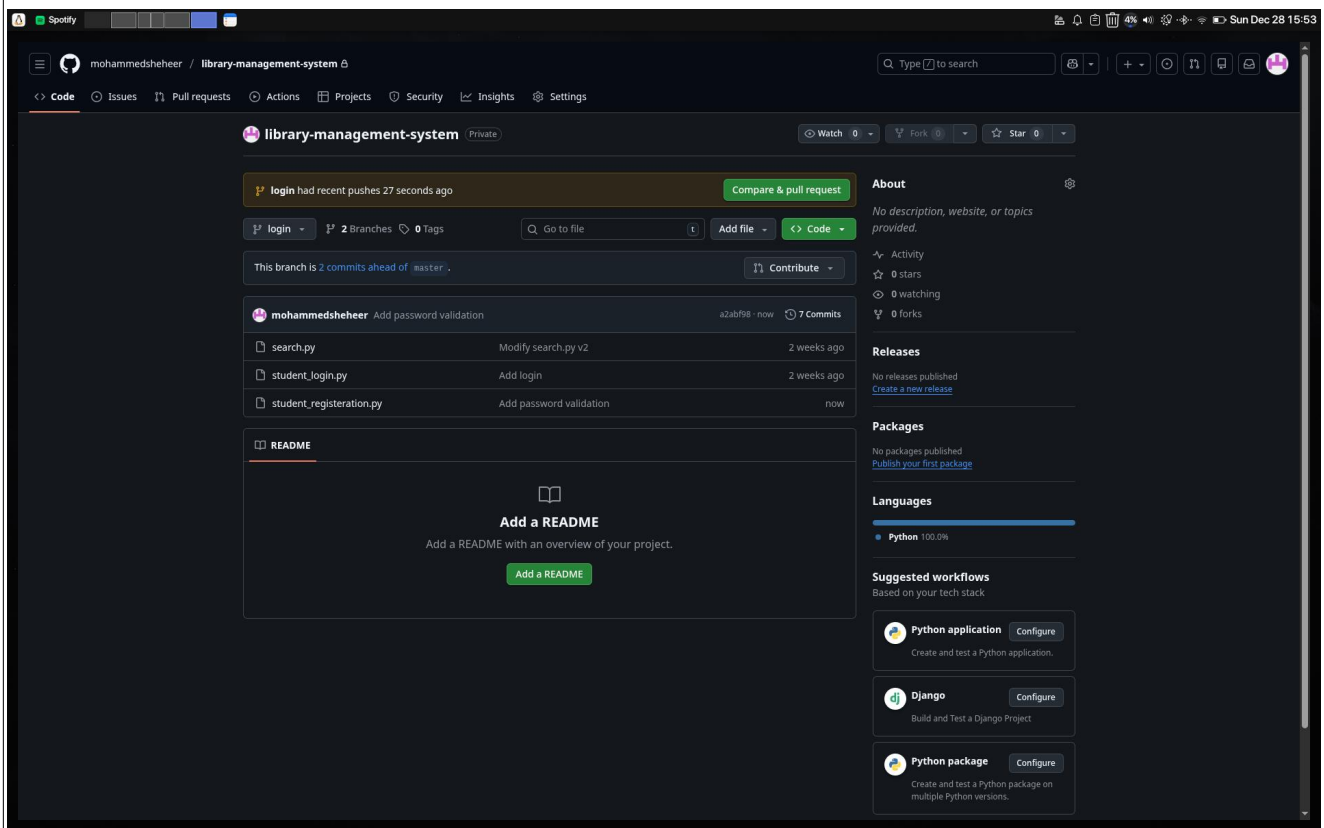
master 0 0 Ln 18, Col 1 Spaces: 4 UTF-8 Python 3.13.11 64-bit

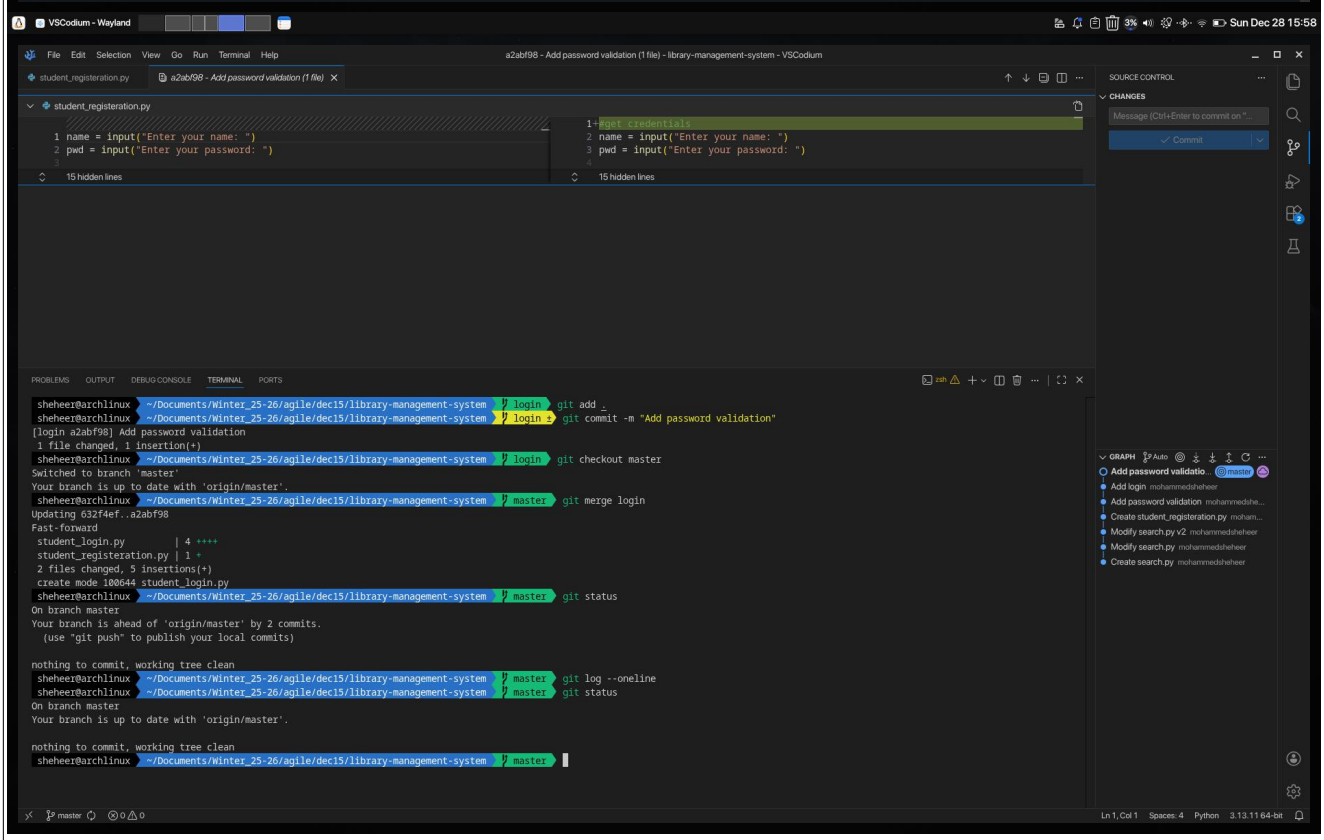
Case Study 3: Feature Development Using Branches





Case Study 4: Merging Feature into Main Code





Spotify Sun Dec 28 16:00

mohammedsheheer / library-management-system

Type / to search

Code Issues Pull requests Actions Projects Security Insights Settings

library-management-system Private

Watch 0 Fork 0 Star 0

master 2 Branches 0 Tags Go to file Add file Code

mohammedsheheer Add password validation a2abf98 · 6 minutes ago 7 Commits

search.py	Modify search.py v2	2 weeks ago
student_login.py	Add login	2 weeks ago
student_registeration.py	Add password validation	6 minutes ago

README

Add a README

Add a README with an overview of your project.

Add a README

About

No description, website, or topics provided.

Activity

0 stars

0 watching

0 forks

Releases

No releases published

Create a new release

Packages

No packages published

Publish your first package

Languages

Python 100.0%

Suggested workflows

Based on your tech stack

Python application Configure

Create and test a Python application.

SLSA Generic generator Configure

Generate SLSA3 provenance for your existing release workflows

Django Configure

Build and Test a Django Project

Spotify Sun Dec 28 16:00

mohammedsheheer / library-management-system

Type / to search

Code Issues Pull requests Actions Projects Security Insights Settings

library-management-system Private

Watch 0 Fork 0 Star 0

login 2 Branches 0 Tags Go to file Add file Code

This branch is up to date with `master`. Contribute

mohammedsheheer Add password validation a2abf98 · 7 minutes ago 7 Commits

search.py	Modify search.py v2	2 weeks ago
student_login.py	Add login	2 weeks ago
student_registeration.py	Add password validation	7 minutes ago

README

Add a README

Add a README with an overview of your project.

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About

No description, website, or topics provided.

Activity

0 stars

0 watching

0 forks

Releases

No releases published

Create a new release

Packages

No packages published

Publish your first package

Languages

Python 100.0%

Suggested workflows

Based on your tech stack

Publish Python Package Configure

Publish a Python Package to PyPI on release.

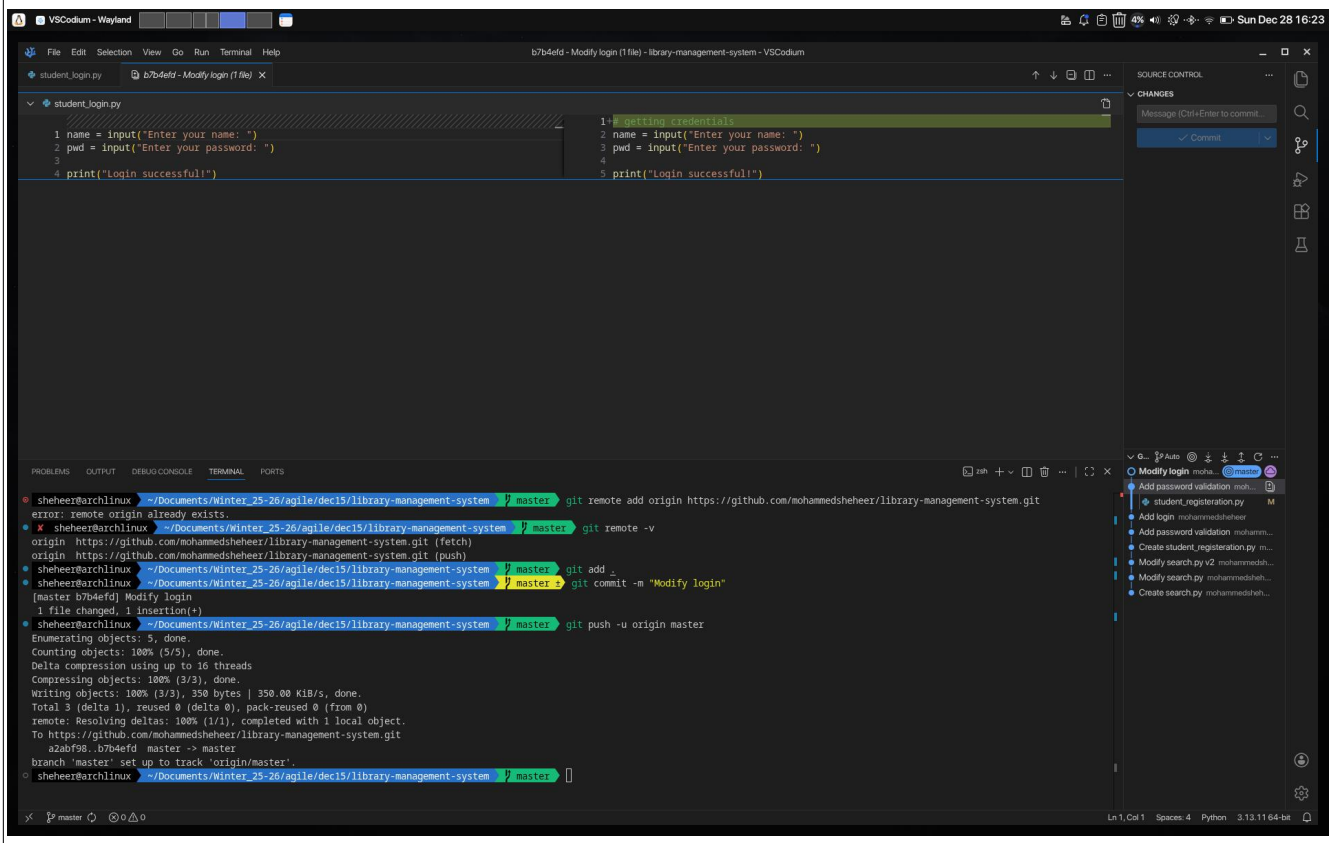
Pylint Configure

Lint a Python application with pylint.

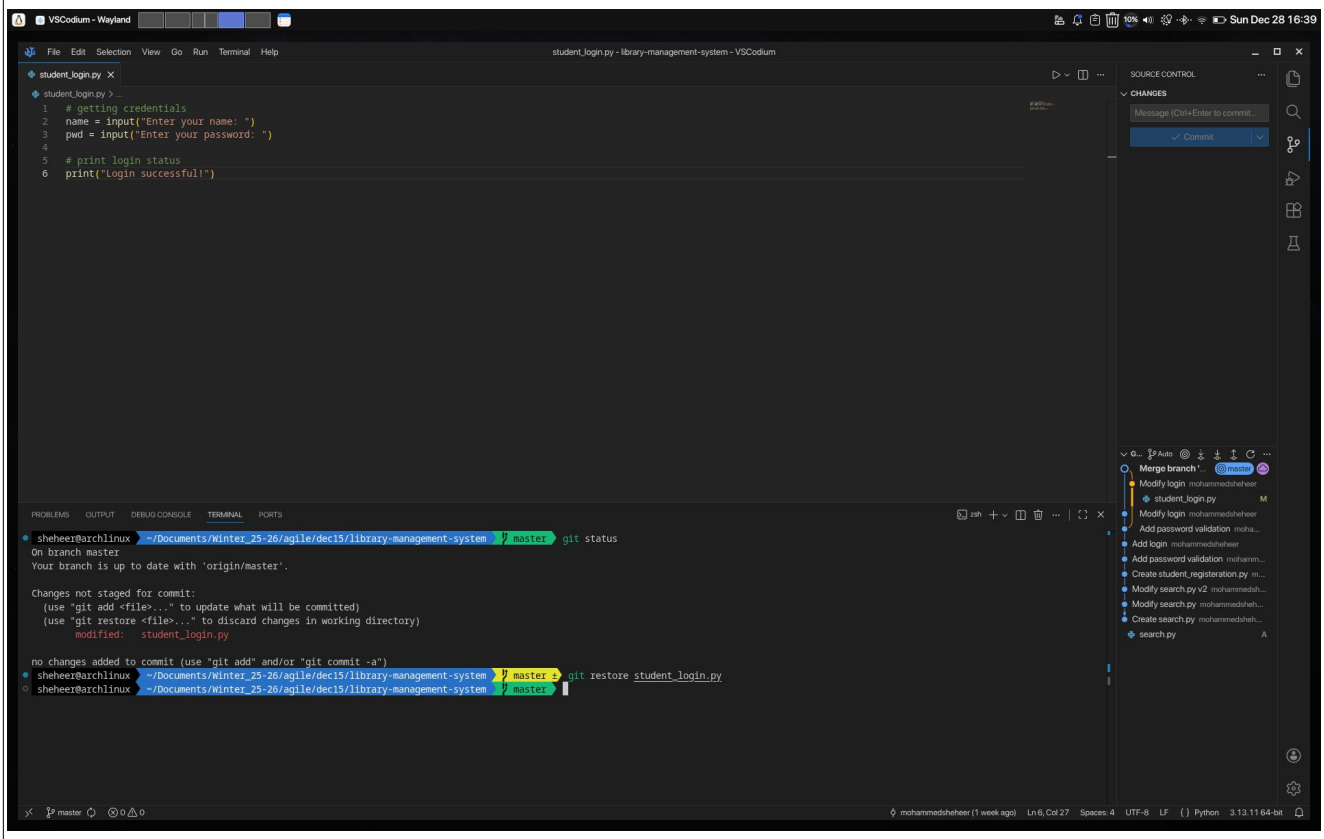
Python package Configure

Create and test a Python package on multiple Python versions.

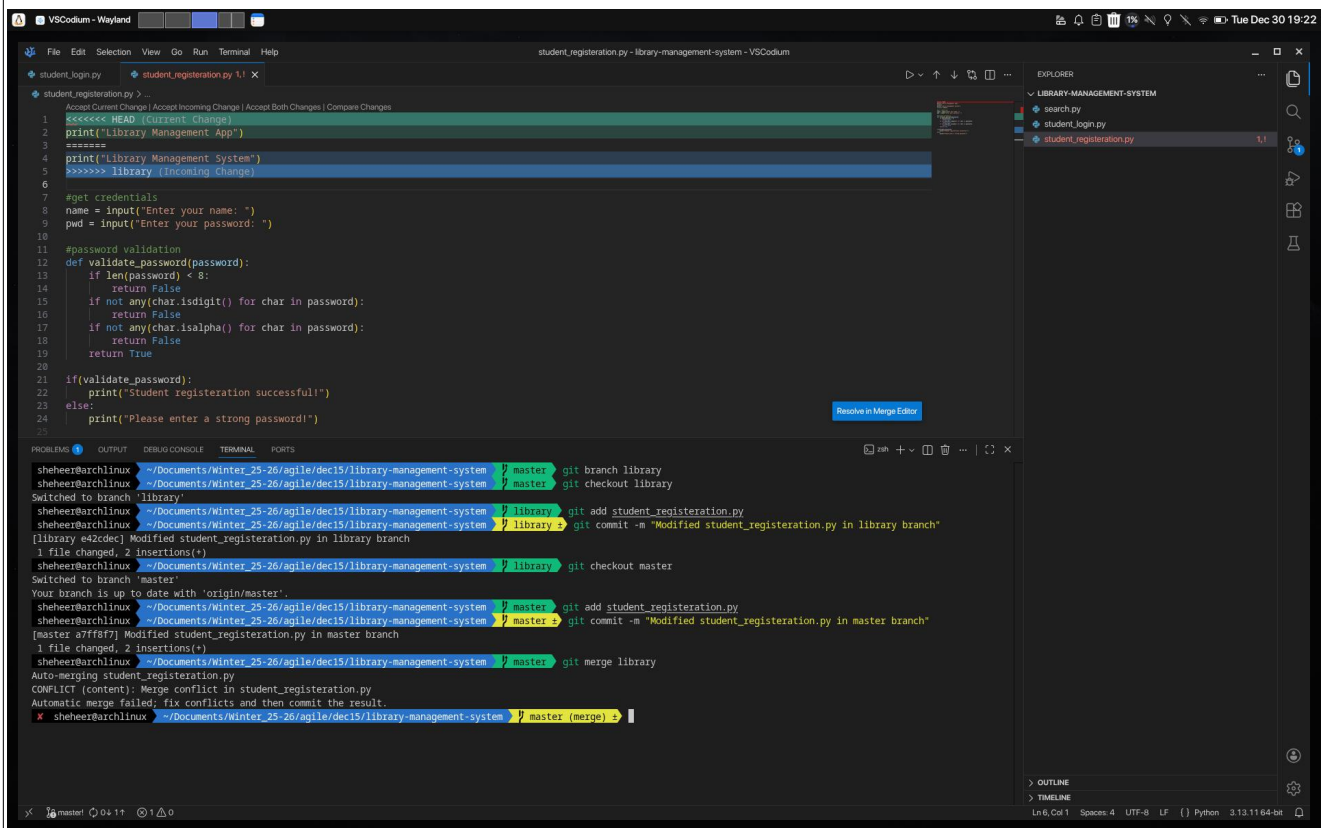
Case Study 5: Working with Remote Repository

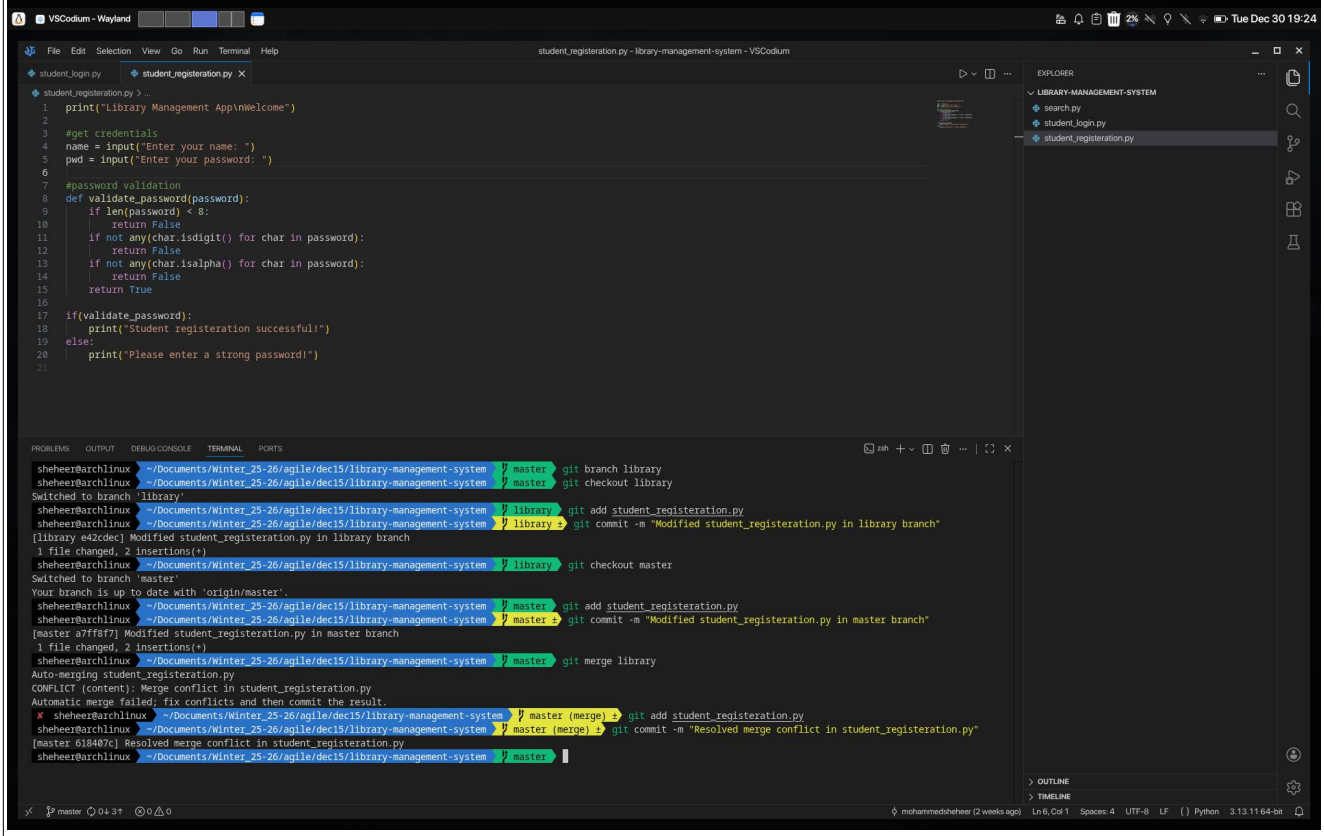


Case Study 6: Undoing Mistakes



Case Study 7: Handling Merge Conflicts





student_registeration.py - library-management-system - VSCodium

student_registeration.py > validate_password

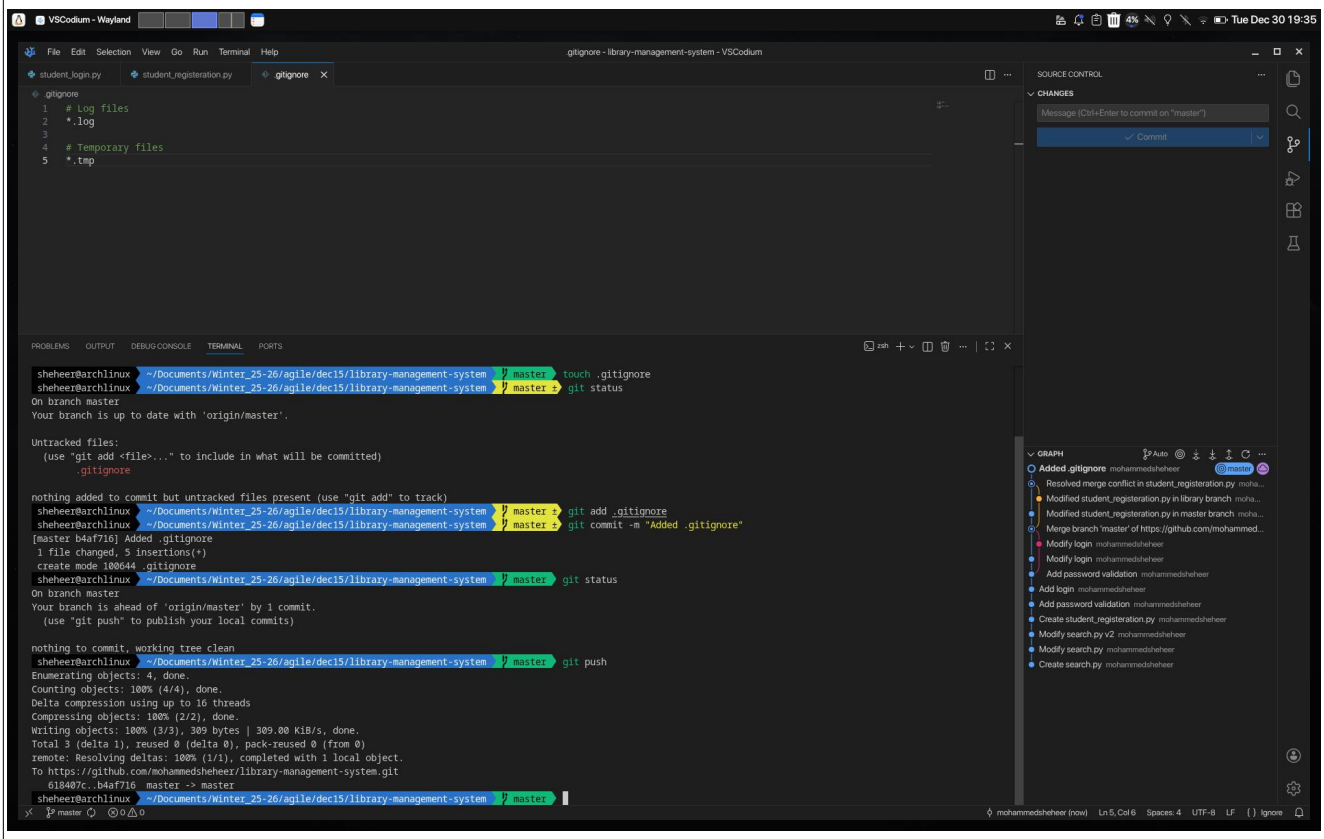
```python
print("Library Management App\nWelcome")

#get credentials
name = input("Enter your name: ")
pwd = input("Enter your password: ")

#password validation
def validate_password(password):
    if len(password) < 8:
        return False
    if not any(char.isdigit() for char in password):
        return False
    if not any(char.isalpha() for char in password):
        return False
    return True
```

```
sheheer@archlinux  ~/Documents/Winter_25-26/agile/dec15/library-management-system  master  git branch library
sheheer@archlinux  ~/Documents/Winter_25-26/agile/dec15/library-management-system  master  git checkout library
Switched to branch 'library'
sheheer@archlinux  ~/Documents/Winter_25-26/agile/dec15/library-management-system  library  git add student_registeration.py
sheheer@archlinux  ~/Documents/Winter_25-26/agile/dec15/library-management-system  library ±  git commit -m "Modified student_registeration.py in library branch"
[library e42cdec] Modified student_registeration.py in library branch
 1 file changed, 2 insertions(+)
sheheer@archlinux  ~/Documents/Winter_25-26/agile/dec15/library-management-system  library  git checkout master
Switched to branch 'master'
Your branch is up to date with 'origin/master'.
sheheer@archlinux  ~/Documents/Winter_25-26/agile/dec15/library-management-system  master  git add student_registeration.py
sheheer@archlinux  ~/Documents/Winter_25-26/agile/dec15/library-management-system  master ±  git commit -m "Modified student_registeration.py in master branch"
[master a7ff8f7] Modified student_registeration.py in master branch
 1 file changed, 2 insertions(+)
sheheer@archlinux  ~/Documents/Winter_25-26/agile/dec15/library-management-system  master  git merge library
Auto-merging student_registeration.py
CONFLICT (content): Merge conflict in student_registeration.py
Automatic merge failed; fix conflicts and then commit the result.
✘ sheheer@archlinux  ~/Documents/Winter_25-26/agile/dec15/library-management-system  master (merge) ±  git add student_registeration.py
sheheer@archlinux  ~/Documents/Winter_25-26/agile/dec15/library-management-system  master (merge) ±  git commit -m "Resolved merge conflict in student_registeration.py"
[master 618407c] Resolved merge conflict in student_registeration.py
sheheer@archlinux  ~/Documents/Winter_25-26/agile/dec15/library-management-system  master  git push
Enumerating objects: 11, done.
Counting objects: 100% (11/11), done.
Delta compression using up to 16 threads
Compressing objects: 100% (9/9), done.
Writing objects: 100% (9/9), 824 bytes | 824.00 KiB/s, done.
Total 9 (delta 7), reused 0 (delta 0), pack-reused 0 (from 0)
remote: Resolving deltas: 100% (7/7), completed with 2 local objects.
To https://github.com/mohammedsheheer/library-management-system.git
   2152aee..618407c  master -> master
sheheer@archlinux  ~/Documents/Winter_25-26/agile/dec15/library-management-system  master
```

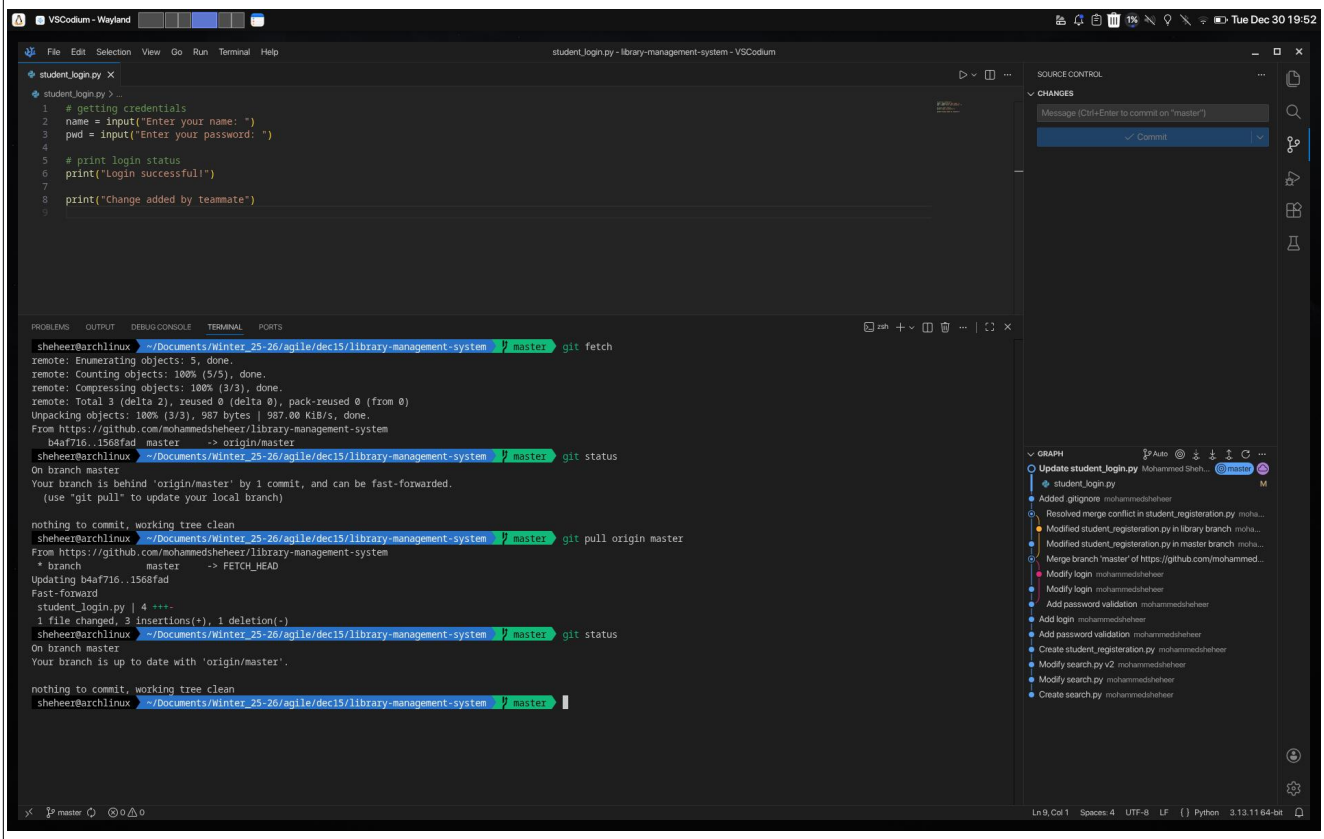
GRAPH

- Resolved merge conflict in student_regi... master
- Modified student_registeration.py in library branch moha...
- Modified student_registeration.py in master branch moha...
- Merge branch 'master' of https://github.com/mohammed...
- Modify login mohammedsheheer
- Modify login mohammedsheheer
- Add password validation mohammedsheheer
- Add login mohammedsheheer
- Add password validation mohammedsheheer
- Create student_registeration.py mohammedsheheer
- Modify search.py v2 mohammedsheheer
- Modify search.py mohammedsheheer
- Create search.py mohammedsheheer

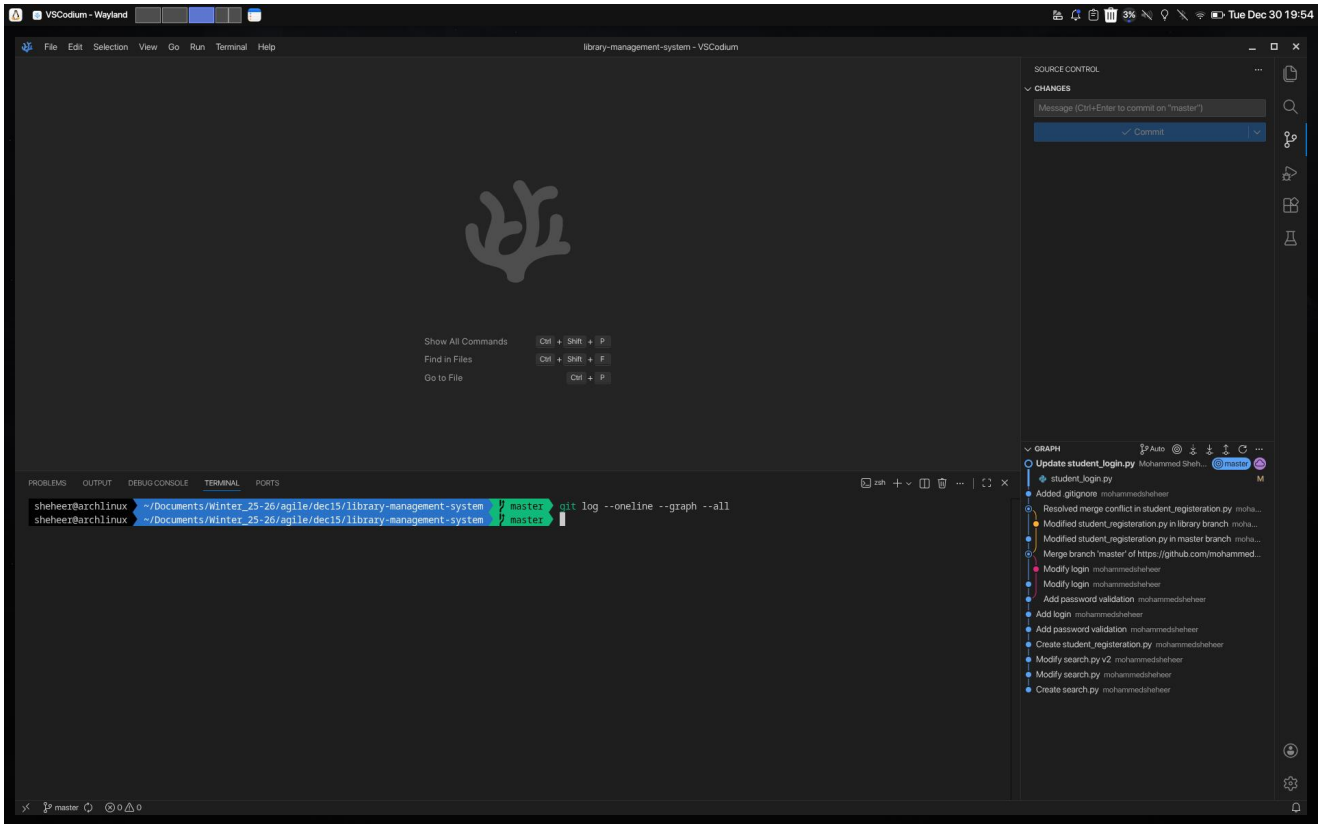
Case Study 8: Ignoring Unwanted Files

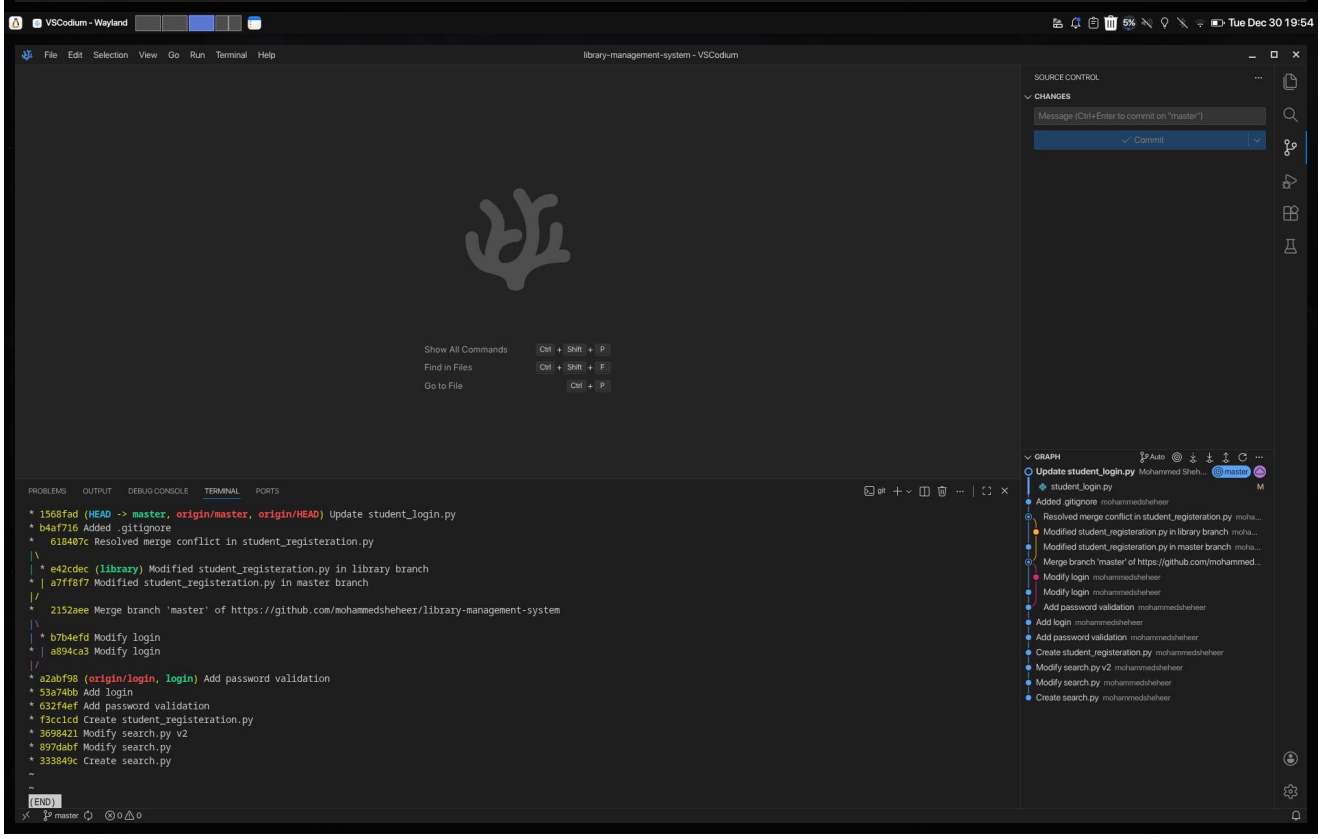


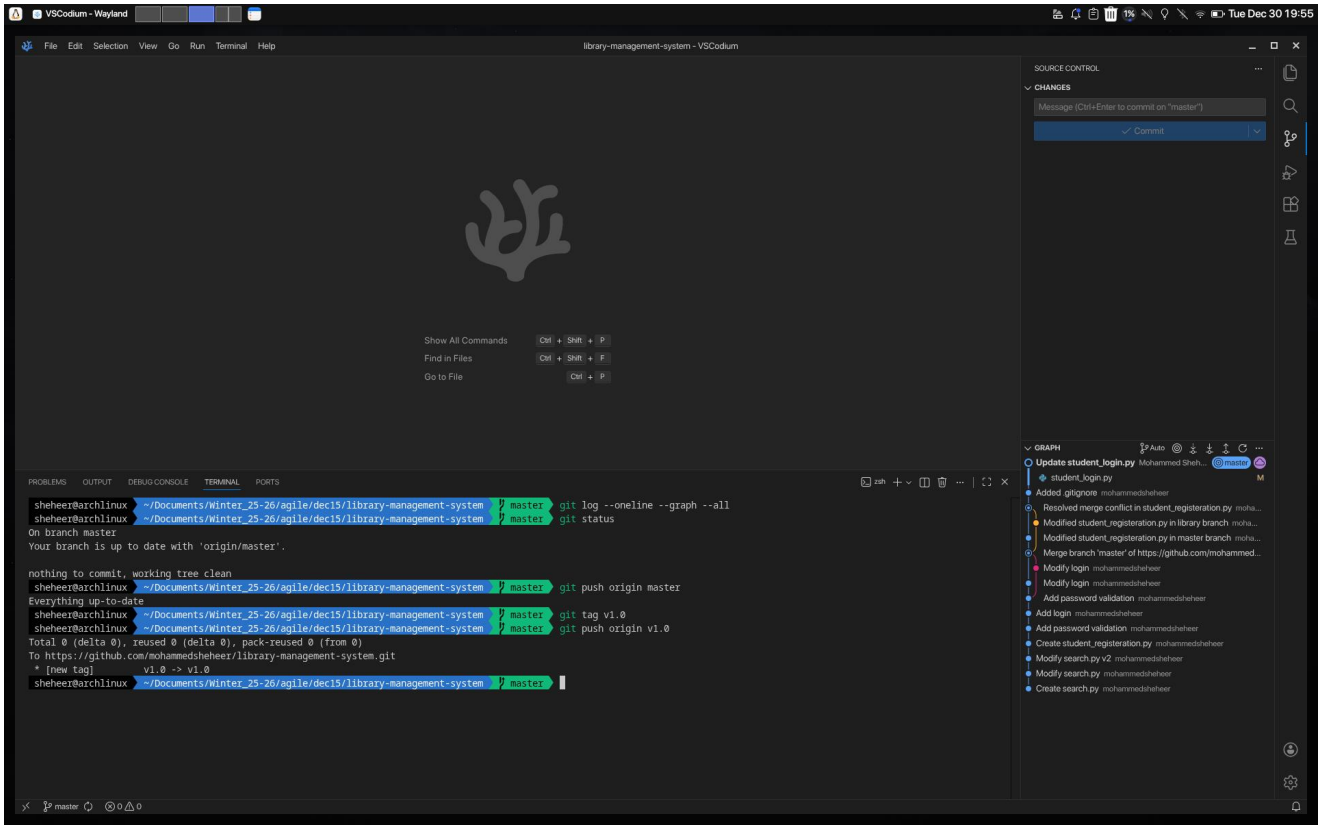
Case Study 9: Collaborating with Team



Case Study 10: Final Project Release






VSCodium - Wayland
Tue Dec 30 19:55
File Edit Selection View Go Run Terminal Help
library-management-system - VSCodium
SOURCE CONTROL
CHANGES
Message (Ctrl+Enter to commit on "master")
Commit
Show All Commands Ctrl + Shift + P
Find in Files Ctrl + Shift + F
Go to File Ctrl + P
GRAPH
Update student_login.py Mohammed Sheh... master
student_login.py M
Added .gitignore mohammedsheheer
Resolved merge conflict in student_registeration.py moha...
Modified student_registeration.py in library branch moha...
Modified student_registeration.py in master branch moha...
Merge branch 'master' of https://github.com/mohammed...
Modify login mohammedsheheer
Modify login mohammedsheheer
Add password validation mohammedsheheer
Add login mohammedsheheer
Add password validation mohammedsheheer
Create student_registeration.py mohammedsheheer
Modify search.py v2 mohammedsheheer
Modify search.py mohammedsheheer
Create search.py mohammedsheheer
PROBLEMS OUTPUT DEBUG CONSOLE TERMINAL PORTS
sheheer@archlinux ~/Documents/Winter_25-26/agile/dec15/library-management-system master git log --oneline --graph --all
sheheer@archlinux ~/Documents/Winter_25-26/agile/dec15/library-management-system master git status
On branch master
Your branch is up to date with 'origin/master'.
nothing to commit, working tree clean
sheheer@archlinux ~/Documents/Winter_25-26/agile/dec15/library-management-system master git push origin master
Everything up-to-date
sheheer@archlinux ~/Documents/Winter_25-26/agile/dec15/library-management-system master git tag v1.0
sheheer@archlinux ~/Documents/Winter_25-26/agile/dec15/library-management-system master git push origin v1.0
Total 0 (delta 0), reused 0 (delta 0), pack-reused 0 (from 0)
To https://github.com/mohammedsheheer/library-management-system.git
* [new tag] v1.0 -> v1.0
sheheer@archlinux ~/Documents/Winter_25-26/agile/dec15/library-management-system master
master 0 0

Class Work

Collaborative Project to work with Git version control

VSCodium - Wayland Mon Jan 05 11:16

File Edit Selection View Go Run Terminal Help balance.py - agile-assignment01 - VSCodium

balance.py > ...

```python
balance = 10000
withdrawAmount = input("Enter the amount to be withdrawn(in Rs): ")

def withdraw(withdrawAmount):
    if(withdrawAmount % 100 == 0 and withdrawAmount > 0 and withdrawAmount < balance):
        balance -= withdrawAmount
        print("Withdrawal successful!")
        print(f"Current balance is {balance}")
```

PROBLEMS OUTPUT DEBUG CONSOLE TERMINAL PORTS

```
sheheer@archlinux ~/Documents/Winter_25-26/agile/jan05/agile-assignment01 main ± git add balance.py
sheheer@archlinux ~/Documents/Winter_25-26/agile/jan05/agile-assignment01 main ± git commit -m "Added balance.py"
[main (root-commit) 3235720] Added balance.py
 1 file changed, 8 insertions(+)
 create mode 100644 balance.py
sheheer@archlinux ~/Documents/Winter_25-26/agile/jan05/agile-assignment01 main git add balance.py
sheheer@archlinux ~/Documents/Winter_25-26/agile/jan05/agile-assignment01 main git status
On branch main
Your branch is based on 'origin/main', but the upstream is gone.
  (use "git branch --unset-upstream" to fixup)

nothing to commit, working tree clean
sheheer@archlinux ~/Documents/Winter_25-26/agile/jan05/agile-assignment01 main git add balance.py
sheheer@archlinux ~/Documents/Winter_25-26/agile/jan05/agile-assignment01 main ± git commit -m "Modified balance.py"
[main 81511b5] Modified balance.py
 1 file changed, 5 insertions(+), 5 deletions(-)
sheheer@archlinux ~/Documents/Winter_25-26/agile/jan05/agile-assignment01 main git status
On branch main
Your branch is up to date with 'origin/main'.

nothing to commit, working tree clean
sheheer@archlinux ~/Documents/Winter_25-26/agile/jan05/agile-assignment01 main git pull origin main
From https://github.com/mohammedsheheer/agile-assignment01
 * branch            main       -> FETCH_HEAD
Updating 81511b5..c4df51c
Fast-forward
 .DS_Store     | Bin 0 -> 6148 bytes
 totalbill.py  |  12 ++++++++++++
 2 files changed, 12 insertions(+)
 create mode 100644 .DS_Store
 create mode 100644 totalbill.py
sheheer@archlinux ~/Documents/Winter_25-26/agile/jan05/agile-assignment01 main
```

SOURCE CONTROL
CHANGES
Message (Ctrl+Enter to commit on "main")
Commit

GRAPH
added totalbill module jashwanth ram pedelli main
Modified balance.py mohammedsheheer
Added balance.py mohammedsheheer

main 0 0 Ln 9, Col 1 Spaces: 4 UTF-8 LF { } Python 3.13.11 64-bit

VSCodium - Wayland Mon Jan 05 13:18

File Edit Selection View Go Run Terminal Help balance.py - agile-assignment01 - VSCodium

balance.py > ...

```python
balance = 10000
withdrawAmount = input("Enter the amount to be withdrawn(in Rs): ")

def withdraw(withdrawAmount):
    if(withdrawAmount % 100 == 0 and withdrawAmount > 0 and withdrawAmount < balance):
        balance -= withdrawAmount
        print("Withdrawal successful!")
        print(f"Current balance is {balance}")
```

PROBLEMS OUTPUT DEBUG CONSOLE TERMINAL PORTS

```
sheheer@archlinux ~/Documents/Winter_25-26/agile/jan05/agile-assignment01 main git pull origin main
From https://github.com/mohammedsheheer/agile-assignment01
 * branch            main       -> FETCH_HEAD
Updating c4df51c..93037fe
Fast-forward
 netSal.py |  9 +++++++++
 tuple.py  | 10 ++++++++++
 2 files changed, 19 insertions(+)
 create mode 100644 netSal.py
 create mode 100644 tuple.py
sheheer@archlinux ~/Documents/Winter_25-26/agile/jan05/agile-assignment01 main git status
On branch main
Your branch is up to date with 'origin/main'.

nothing to commit, working tree clean
sheheer@archlinux ~/Documents/Winter_25-26/agile/jan05/agile-assignment01 main
```

SOURCE CONTROL
CHANGES
Message (Ctrl+Enter to commit on "main")
Commit

GRAPH
tuple.py swetha290605 main
added NETSallary prasad-1807
added totalbill module jashwanth ram pedelli
Modified balance.py mohammedsheheer
Added balance.py mohammedsheheer

main 0 0 Ln 9, Col 1 Spaces: 4 UTF-8 LF { } Python 3.13.11 64-bit

VSCodium - Wayland Mon Jan 05 13:19

File Edit Selection View Go Run Terminal Help balance.py - agile-assignment01 - VSCodium

balance.py

```python
balance = 10000
withdrawAmount = input("Enter the amount to be withdrawn(in Rs): ")

def withdraw(withdrawAmount):
    if(withdrawAmount % 100 == 0 and withdrawAmount > 0 and withdrawAmount < balance):
        balance -= withdrawAmount
        print("Withdrawal successful!")
        print(f"Current balance is {balance}")
```

PROBLEMS OUTPUT DEBUG CONSOLE TERMINAL PORTS

```
sheheer@archlinux  ~/Documents/Winter_25-26/agile/jan05/agile-assignment01  main  git log --graph --all
```

SOURCE CONTROL
CHANGES
Message (Ctrl+Enter to commit on "main")
Commit

GRAPH
tuple.py swetha290605 main
added NETSallary prasad-1807
added totalbill module jashwanth ram pedelli
Modified balance.py mohammedsheheer
Added balance.py mohammedsheheer

main 0 0 Ln 9, Col 1 Spaces: 4 UTF-8 LF Python 3.13.11 64-bit

VSCodium - Wayland Mon Jan 05 13:19

File Edit Selection View Go Run Terminal Help balance.py - agile-assignment01 - VSCodium

balance.py

```python
balance = 10000
withdrawAmount = input("Enter the amount to be withdrawn(in Rs): ")

def withdraw(withdrawAmount):
    if(withdrawAmount % 100 == 0 and withdrawAmount > 0 and withdrawAmount < balance):
        balance -= withdrawAmount
        print("Withdrawal successful!")
        print(f"Current balance is {balance}")
```

PROBLEMS OUTPUT DEBUG CONSOLE TERMINAL PORTS

```
* commit 93037fe2ca3f0a5cb2c3cac36d494d1ade421c1b (HEAD -> main, origin/main, origin/HEAD)
| Author: swetha290605 <sevuganswetha@gmail.com>
| Date:   Mon Jan 5 12:08:23 2026 +0530
|
|     tuple.py
|
* commit 079364538286b2724570279f0ffc8e3b8902862f
| Author: prasad-1807 <sigatapu.siva2023@vitstudent.ac.in>
| Date:   Mon Jan 5 11:22:45 2026 +0530
|
|     added NETSallary
|
* commit c4df51c872e2b26bd944318d24c58fbeb6585f4d
| Author: jashwanth ram pedelli <jashwanthrampedelli@jashwanths-MacBook-Air.local>
| Date:   Mon Jan 5 11:14:02 2026 +0530
|
|     added totalbill module

* commit 81511b5d9c45aaff8bfc8d687f69c76649ea0b6d
| Author: mohammedsheheer <mohammed.sheheer2@gmail.com>
| Date:   Mon Jan 5 10:44:08 2026 +0530
|
|     Modified balance.py
|
* commit 3235720fb3361e1a18398ad3991315aa5f21641f
  Author: mohammedsheheer <mohammed.sheheer2@gmail.com>
  Date:   Mon Jan 5 10:42:19 2026 +0530

      Added balance.py
(END)
```

SOURCE CONTROL
CHANGES
Message (Ctrl+Enter to commit on "main")
Commit

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